Monetary Policy Shocks, Large Language Models and Cross-Validation*

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Abstract

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We identify monetary policy shocks using the narrative approach, incorporating textual information from FOMC documents via large language model embeddings. This procedure poses challenges due to the limited sample size and high dimensionality of the variable set. To address this, we implement out-of-sample cross-validation procedures, which we prove are essential for identification of the monetary shocks. Furthermore, we show which information is captured within the embeddings and how the generated shocks produce macroeconomic responses consistent with theory.

Keywords: Monetary policy shocks, Narrative identification, Large language models, Cross-validation, FOMC communications.

JEL Codes: E52, C45, C53, E58.

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1 Introduction

Understanding the effects of monetary policy is essential to both economic theory and policymaking. Central bank decisions profoundly shape output, inflation, employment, and financial conditions, influencing the business cycle and long-term growth. However, isolating their causal impact from systematic responses to economic conditions and from private sector expectations has remained a fundamental challenge over decades of empirical research. Despite important advances, key puzzles persist regarding the timing and magnitude of policy transmission.

Our paper relies on the Narrative Approach, which separates changes in the policy rate into systematic and nonsystematic components. The systematic component reflects policy adjustments in response to current and expected economic conditions, which accounts for the central bank information set, and represents the endogenous part of monetary policy. The nonsystematic component consists of policy rate changes that are not explained by the systematic part and is interpreted as the exogenous variation, or shock. The causal effects of monetary policy are related to this nonsystematic component.

A reliable proxy for the central bank's information set, which underlies the systematic component, is critical under this strategy. We use forecasts and textual data from the Federal Reserve, relying on documents prepared by the Fed staff for use in the Federal Open Market Committee (FOMC) meetings. These documents provide a clear view of the thinking of policymakers immediately prior to each meeting. In particular, we use the Greenbook and the Beige Book. These materials contain not only relevant forecast data for the main macroeconomic variables used in setting the policy rate (the federal funds rate), but also detailed discussions of current economic conditions.

Separating the effects of the systematic and nonsystematic components poses significant challenges, primarily due to the limited sample size and the high dimensionality of the variable set. This issue can be addressed by employing machine learning techniques. However, the interaction between a small sample and a large number of predictors can cause overfitting, which undermines the model's ability to generalize to new data. To our knowledge, this paper is the first to address this issue by evaluating out-of-sample the residual of the models. We propose two approaches: the Leave-One-Out Cross-Validation (LOOCV) and the Expanding Window (EW) procedure. This framework is designed to account for the challenges inherent in identifying monetary policy shocks and represents a new contribution to the literature which we argue is not only a diagnostic tool, but also a demand for proper identification.

We use Large Language Models to represent the textual information. LLMs provide a more comprehensive analysis of semantics and are capable of capturing meaning across

longer textual spans and outperforming simple Natural Language Processing methods. We embed the full text using an LLM and incorporate these representations directly into the Fed's information set instead of using a single dimension. This eliminates the information loss inherent in choosing a single summary dimension, at the cost of working with high-dimensional embeddings. We also provide evidence on what can be learned from our LLM embeddings, showing that they capture important features embedded in the language of the FOMC documents and that their use allows us to build a broader measure of the fed information set compared to prior works in the literature.

Our shock identification progresses in two steps. First, we select the LLM embedding series with the strongest out-of-sample predictive performance for interest rate changes. Second, conditional on this choice, we select the machine learning estimator on the same criterion. Both steps are governed by a clearly defined out-of-sample metric, a novelty in the literature. We then assess whether the identified shocks produce macroeconomic responses consistent with theory, a standard validation exercise in the monetary policy literature. To assess this, we estimate dynamic causal effects using local projections, with our shock series serving as an external instrument for the policy rate to isolate exogenous variation in monetary policy.

Our findings can be summarized as follows. Cross-validation is consequential for shock identification: replicating prior work, we show that the choice of validation procedure is sufficient to reverse the sign of estimated monetary policy effects on the consumer price index (CPI). LLM embeddings generate a broader measure of the Fed's information set, a result that is not mechanical, as the embeddings capture macroeconomic content embedded in FOMC language and display stronger predictive power for the policy rate than the numerical forecasts contained in the same documents. Our preferred shock series produces the expected causal responses: a persistent increase in the policy rate, a decline in industrial production, and a fall in the price level within six months, the latter, substantially narrowing the gap between narrative and high-frequency identification strategies.

2 Literature and Contributions

Understanding the macroeconomic effects of monetary policy requires isolating exogenous variation in the policy instrument. The recursive approach of Christiano et al. (1996) identifies shocks by ordering the policy instrument last in a VAR, but famously generates a "price puzzle" where prices rise following a policy tightening.[1] The high-frequency

[1]The recursive approach has also been extended using sign restrictions to aid identification (e.g., Uhlig (2005)) and by expanding the set of variables in the VAR framework, as in FAVAR models (e.g., Bernanke et al. (2005)).

approach of Gürkaynak et al. (2005), Gertler and Karadi (2015), Nakamura and Steinsson (2018), Jarociński and Karadi (2020), Miranda-Agrippino and Ricco (2021), Bauer and Swanson (2022), Acosta (2023), Bauer and Swanson (2023), Swanson and Jayawickrema (2023), and Jarociński and Karadi (2025), among others, uses variations in asset prices (mainly interest rate futures contracts) within a narrow time window around FOMC announcements to estimate monetary policy shocks.

The narrative approach, pioneered by Romer and Romer (2004) and further applied by Cloyne and Hürtgen (2016), Holm et al. (2021), and Aruoba and Drechsel (2024), among others, uses narrative records from FOMC meetings to infer when the central bank changes the policy rate for exogenous reasons, that is, changes not contaminated by systematic policy responses to macroeconomic conditions. We follow this strategy but depart from existing implementations: while prior work relies exclusively on in-sample estimation to validate residuals, we introduce systematic out-of-sample evaluation via nested cross-validation.

This paper also contributes to the emerging literature that uses text analysis to identify monetary policy shocks Handlan (2022), Acosta (2023), Aruoba and Drechsel (2024) among others. Handlan (2022) employs neural networks to translate FOMC statement texts into predicted changes in federal funds rate expectations, capturing expectation shifts not directly tied to the policy action. She further refines this measure using alternative FOMC texts to account for the informational content of the announcements. Aruoba and Drechsel (2024) apply machine learning and dictionary methods to construct a cleaned policy shock series based on central bank texts. Their series, spanning 1982 to 2008, produces macroeconomic responses consistent with theory, though it exhibits a mild "price puzzle". Acosta (2023) uses newspaper coverage of FOMC meetings to build high-frequency measures of forecast revisions. This method captures expectation shifts linked specifically to policy announcements and yields shocks that align with standard theoretical predictions. We use LLMs to process textual data contained in the FOMC documents which give us clearly advantages over the traditional dictionary methods. In brief, LLMs provide a larger set of information due to three key advantages: are multi-purpose, are context-sensitive and have superior performance on both classification and regressions tasks.

Finally, our evaluation framework bridges econometrics and machine learning literatures on cross-validation. We trace foundational contributions such as Allen (1974) and Stone (1974), who proposed out-of-sample metrics to assess model performance. We also incorporate more recent insights from the Machine Learning literature on overfitting (Hawkins, 2004; Cawley and Talbot, 2010) and data leakage (Cawley and Talbot, 2010; Kaufman et al., 2012). In addition, we discuss the specific challenges of cross-validation in time series contexts, as explored in Bergmeir et al. (2018), Cerqueira et al. (2020), and

Cerqueira et al. (2022). More specifically, we discuss nested cross-validation techniques, which have been thoroughly explored outside of the residual estimation task, Stone (1974), Varma and Simon (2006), Filzmoser et al. (2009), Cawley and Talbot (2010), Krstajic et al. (2014), Parvandeh et al. (2020) and Wainer and Cawley (2021). To our knowledge, ours is the first paper to apply these techniques to monetary policy shock identification.

3 Cross-validating Monetary Policy Shocks

3.1 Revisiting the Narrative Approach

The narrative approach to monetary policy identification comes from a simple premise: given the information set available to the Federal Reserve at the time of each decision, the residual variation in interest rate changes can be interpreted as an exogenous policy shock. Since the seminal contribution of Romer and Romer (2004), this approach has been operationalized by projecting interest rate changes onto observable measures of the Fed's information set.[2] Formally, central banks adjust the short-term interest rate in response to both current economic conditions and forward-looking considerations, following a reaction function of the form:

$$\Delta i_t = \Gamma(X_t) + \varepsilon_t \tag{1}$$

where Δi_t denotes the change in the short-term interest rate at each FOMC meeting, X_t is a proxy for the information set available to policymakers just prior to the meeting, and $\Gamma(\cdot)$ is a general function capturing the systematic component of monetary policy. The residual ε_t represents the nonsystematic component, exogenous monetary policy variations that are, by construction, orthogonal to $\Gamma(X_t)$.

Monetary policy surprises then arise from two distinct sources: (i) a pure exogenous shock ε_t, and (ii) a Fed Information Effect (FIE), which is a result of different information sets just prior to a meeting by the central bank and the private sector. We provide a clear definition of what is the Fed Information Effect (FIE): any information contained in FOMC announcements that relates to either X_t or $\Gamma(\cdot)$ and is capable of surprising the public. It is important to note that this is also compatible with the theoretical model of central bank asymmetric information developed by Miranda-Agrippino and Ricco (2021). Under the narrative approach, this definition separates our identification strategy from the critique advanced by Bauer and Swanson (2023), namely the preferred Fed response to news (FRN) channel.[3]

The choice of proxy X_t is central to the narrative approach. Since Romer and Romer (2004), the Greenbook forecasts have served as the benchmark: material prepared by

[2] As mentioned before, other strategies to identify monetary policy shocks exist, such as the recursive and high-frequency approaches. While the narrative approach allows for the measurement of shocks in the 1980s, the high-frequency identification approach makes it possible to identify shocks during the zero lower bound period, which became a defining feature of monetary policy in the aftermath of the Global Financial Crisis.

[3] There are timing issues between the release of the Greenbook and the meeting itself, but Aruoba and Drechsel (2024) provide evidence that this does not raise concerns about the information flow between the two.

Fed staff six days before each meeting, containing nowcasts and projections for a range of macroeconomic variables up to four quarters ahead. A complementary document, the Beige Book, discusses economic conditions across Federal Reserve districts and, unlike the Greenbook, which is publicly available 5 years after the meeting, is publicly available prior to every FOMC meeting. Following Aruoba and Drechsel (2024), the information set has been expanded beyond the numerical forecasts of Romer and Romer (2004) by also incorporating the textual content of both documents.[4] [5]

The function $\Gamma(\cdot)$ is a second dimension which has been improved. Rather than restricting the systematic component to an Ordinary Least Squares, literature has expanded to add regularized linear models . This flexibility allows the model to capture more precisely higher order and interaction effects in the Fed's reaction function that a linear projection would miss.

The exogenous monetary policy shock is then the estimated residual $\hat{\varepsilon}_t$ from equation 1. For this shock series to support valid causal inference, it must satisfy two properties: exogeneity and unpredictability. The former follows by construction, since controlling for the systematic component $\Gamma(X_t)$ renders the residuals orthogonal to it. The latter is ensured by incorporating forward-looking forecasts into X_t, thereby purging the anticipatory behavior inherent in policy decisions.

3.2 The importance of Cross-Validation

Traditional econometric work on the identification of monetary policy often relies on OLS regressions, but this technique has two limitations in our setting where both the amount of forecasts and text variables is growing. First, OLS becomes unreliable in high-dimensional environments, either when the number of regressors (p) exceeds the number of observations (n), or in sequences where both are increasing, but at a constant rate. In these cases, the OLS estimator is infeasible or unstable, and predicted residuals can be highly sensitive to individual observations.

Second, when the object of interest is a shock constructed as an in-sample residual, OLS implies that the residual variance is weakly non-increasing as the set of controls expands. Consequently, enlarging the information set, for example by adding lags or additional predictors intended to better approximate the central bank's information set,

[4]It is beyond the scope of this paper to provide evidence on the forecasting quality of the Greenbook. Romer and Romer (2000) and Miranda-Agrippino and Ricco (2021) provide evidence of its superior forecasting ability; Although Nakamura and Steinsson (2018) finds evidence in favor of FIE, they argue that neither the Fed nor the private sector holds a clear advantage; and Bauer and Swanson (2023) finds accuracy broadly comparable to the Blue Chip survey of professional forecasters.

[5]We do not use the information contained in the FOMC minutes or transcripts, as our objective is to limit the information set to what is available just prior to the meeting. These two documents contain clear information about the policy decision itself.

will mechanically leave the estimated monetary policy shock unchanged or smaller, even if the additional variables do not improve predictive performance. To address these problems, we use machine learning (ML) methods and an extensive cross-validation strategy designed to limit overfitting when constructing the residual.

This issue is widely discussed in the literature on leave-one-out regression and overfitting: predictions can become overly sensitive to the training sample, particularly when the same observation contributes to both model fitting and evaluation. To mitigate this concern, we implement nested cross-validation as proposed by Stone (1974). The dataset is first split into training and test samples; within the training sample, an inner cross-validation loop is used for model selection and hyperparameter tuning, while the test sample is reserved exclusively for evaluation. In our application, the residual is computed out of sample as the difference between the model's prediction for the test observation and its realized value.

The choice of nested cross-validation is not without critique. Wainer and Cawley (2021) argue that this substantially more computationally intensive procedure can be overly conservative and that, in many empirical applications with machine learning estimators, standard cross-validation delivers comparable performance assessment. Nevertheless, in our application the object of interest is not predictive accuracy per se, but a sequence of residuals that is subsequently interpreted as a shock. In this setting, even modest overfitting or tuning-induced optimism can translate into nontrivial distortions in the estimated residual process, so the additional separation between tuning and evaluation provided by nested cross-validation can be consequential. We show below that, for residual estimation in this environment, nested cross-validation yields meaningful gains relative to simpler procedures.

We implement two variants of this approach:

- *Leave-One-Out (LOOCV):* For each observation t, we exclude it from the training sample and estimate the model on the remaining observations. Repeating this for all t yields a full sequence of out-of-sample predicted residuals. This procedure mitigates overfitting and reduces leverage effects relative to in-sample residuals, but it does not fully rule out leakage in time-series settings because the training sample for a given t may include observations dated after t, so the prediction can implicitly use future information (Cerqueira et al., 2020).

- *Expanding-window (EW) forecasting:* We initialize the training sample with the first i time-ordered observations and generate a one-step-ahead prediction for observation $i + 1$. The training window is then expanded by one observation at a time, producing one-step-ahead predictions and residuals for the remaining $n - i$ observations. This scheme respects the temporal ordering and rules out leakage

by construction, at the cost of relying on relatively short training samples early in the evaluation period and therefore using the available data less efficiently for estimating low-frequency components of the relationship.

3.3 Econometric Framework

In this section, we formalize the three evaluation strategies mentioned above, in-sample, leave-one-out cross-validation (LOOCV), and expanding-window (EW), along with their key shortcomings and relative qualities for time series residual construction. Anticipating the results, we show that in-sample estimators suffer from multiple biases, and while LOOCV appears data-efficient relative to EW, lookahead bias makes it less reliable for real-time forecasting. The decomposition below reveals why only causal expanding-window evaluation delivers unbiased estimates of true policy predictability.

Let $\{(X_t, \Delta i_t)\}_{t=1}^{T}$ be a time-ordered sample, where X_t denotes predictors available at time t and Δi_t is the explanatory variable. Let $\widehat{\Gamma}^{S}(\cdot)$ denote the prediction rule trained on an index set $S \subset \{1, \ldots, T\}$. For any t, define the forecast error produced by a rule trained on S as

$$\widehat{e}_t(S) \;\equiv\; \Delta i_t - \widehat{\Gamma}^{S}(X_t), \qquad \widehat{\mathrm{Err}}(S) \;\equiv\; \frac{1}{|S'|}\sum_{t\in S'} \widehat{e}_t(S)^2,$$

where S' is the evaluation. We focus on squared error loss.

(1) In-sample (IS) error. Train once on the full sample $S^{IS} = \{1, \ldots, T\}$ and evaluate on the same observations, $S' = S$:

$$\widehat{\mathrm{Err}}^{\,IS} \;\equiv\; \frac{1}{T}\sum_{t=1}^{T}\Big(\Delta i_t - \widehat{\Gamma}^{\{1:T\}}(X_t)\Big)^2. \tag{2}$$

Two problems appear on in-sample estimation, lookahead bias, which is explored on (2) and optimism. Since the same data is used for training and evaluation, $\widehat{\mathrm{Err}}^{\,IS}$ is typically optimistic for out-of-sample performance. Under squared error loss, the expected out-of-sample prediction error equals the expected in-sample error plus a covariance penalty (Efron (2004)):

$$\mathbb{E}[\mathrm{Err}] = \mathbb{E}[\hat{\mathrm{Err}}^{\mathrm{IS}}] + \frac{2}{T}\sum_{t=1}^{T}\mathrm{cov}\big(\widehat{\Gamma}^{\{1:T\}}(X_t),\, \Delta i_t\big), \tag{3}$$

so whenever the average covariance term is positive, one has

$$\mathbb{E}[\widehat{\mathrm{Err}}^{\,IS}] \;<\; \mathbb{E}[\mathrm{Err}].$$

(2) Leave-one-out cross-validation (LOOCV). For each t, train on $S^{(-t)} = \{1, \ldots, T\} \setminus \{t\}$ and evaluate at the left-out point:

$$\widehat{\mathrm{Err}}^{LOOCV} \equiv \frac{1}{T} \sum_{t=1}^{T} \Big(\Delta i_t - \widehat{\Gamma}^{\{1:T\}\setminus\{t\}}(X_t) \Big)^2. \tag{4}$$

In a time series, $\widehat{\Gamma}^{\{1:T\}\setminus\{t\}}$ uses observations from times $s > t$ when predicting at time t, which is infeasible for real-time forecasting and can make $\widehat{\mathrm{Err}}^{LOOCV}$ artificially small relative to a feasible forecast evaluation. To observe this, define two sigma-algebras for each t and the following oracle predictors:

$$\mathcal{F}_{t-1} = \sigma\big(\{X_s, \Delta i_s\}_{s<t}\big), \qquad \mathcal{F}_{-t} = \sigma\big(\{X_s, \Delta i_s\}_{s\neq t}\big).$$

$$f_t^* = \mathbb{E}[\Delta i_t \mid \mathcal{F}_{t-1}] \quad \text{(Expanding-Window Oracle)},$$

$$g_t^* = \mathbb{E}[\Delta i_t \mid \mathcal{F}_{-t}] \quad \text{(LOOCV Oracle)}.$$

Because $\mathcal{F}_{t-1} \subseteq \mathcal{F}_{-t}$, the orthogonality of L^2 projections yields the Pythagorean decomposition: [6]

$$\underbrace{\mathbb{E}\big[(\Delta i_t - f_t^*)^2\big]}_{\text{EW oracle MSE}} = \underbrace{\mathbb{E}\big[(\Delta i_t - g_t^*)^2\big]}_{\text{LOOCV oracle MSE}} + \underbrace{\mathbb{E}\big[(g_t^* - f_t^*)^2\big]}_{\text{leakage gap} \geq 0}. \tag{5}$$

The leakage gap admits the variance representation

$$\mathbb{E}\big[(g_t^* - f_t^*)^2\big] = \mathbb{E}(\mathrm{Var}\Big(\mathbb{E}[\Delta i_t \mid \mathcal{F}_{-t}] \;\Big|\; \mathcal{F}_{t-1}\Big)) \geq 0,$$

Intuitively, the sigma-algebras represent the information available for prediction. $\mathcal{F}_{t-1}$ contains only past data, what policymakers would have had at time t. $\mathcal{F}_{-t}$ includes future observations ($s > t$), which leak infeasible information into LOOCV predictions. The decomposition shows that this leakage strictly improves LOOCV performance relative to what any causal forecaster could achieve using only past data. Hence LOOCV is *weakly optimistic* relative to causal expanding-window evaluation at the oracle level, with bias increasing in temporal dependence.

(3) Expanding-window error. For forecasting, a standard feasible evaluation is expanding-window testing: for each t in an evaluation range $\{t_0, \ldots, T\}$, train only on the past

[6] The cross-term vanishes because the error of the informed oracle ($\Delta i_t - g_t^*$) is, by definition, "pure noise" relative to the information it possesses. Since the leakage gap ($g_t^* - f_t^*$) is built entirely from that same information, the two terms are uncorrelated. Geometrically, this means the prediction error is always perpendicular to the information used to create it, allowing for a clean Pythagorean decomposition of the total MSE.

$S_t^{EW} = \{1, \ldots, t-1\}$ and evaluate at t:

$$\widehat{\text{Err}}^{EW} \equiv \frac{1}{T - t_0 + 1} \sum_{t=t_0}^{T} \left(\Delta i_t - \widehat{\Gamma}^{\{1:t-1\}}(X_t) \right)^2. \tag{6}$$

Relative to (4), each term in (6) is computed from a smaller training set, especially early in the sample. Moreover, because the model is re-estimated many times and each training set can be small, $\widehat{\text{Err}}^{EW}$ can be more variable in finite samples, even though it targets the relevant feasible forecasting risk.

3.4 Cross-Validation: A case study

To illustrate the impact which cross-validation can have, we replicate the results on Romer and Romer (2004) and Aruoba and Drechsel (2024). We restrict our analysis to the period from October 1982 to October 2008, thereby avoiding both the zero lower bound (ZLB) period and the era preceding modern monetary policy. This yields 210 FOMC meetings for regression.

For Romer and Romer (2004), we use the forecasts ranging from the previous quarter to two quarters ahead for Real Gross Domestic Product, Gross Domestic Product Price Index and Unemployment (only for the current quarter). Additionally, they include their first differences, ranging from the previous quarter to two quarters ahead, resulting in a total of 18 regressors. While for Aruoba and Drechsel (2024), the forecasts are extended using macroeconomic variables ranging from the previous quarter to four quarters ahead, including: Headline Consumer Price Index, Real Personal Consumption Expenditures, Real Business Fixed Investment, Real Residential Investment , Real Federal Government Consumption and Gross Investment, Real State and Local Government Consumption and Gross Investment, Nominal Gross Domestic Product, Housing and Industrial Production Index. Additionally, they include their first differences, ranging from the previous quarter to three quarters ahead.

Table 1 reports the R^2 statistics for each machine learning estimator across three evaluation criteria. Comparing the naïve in-sample R^2 against the expanding window and LOOCV metrics directly illustrates the consequences of overfitting and data leakage: specifications that show substantial gains in-sample frequently deliver similar or worse predictive accuracy out-of-sample. This deterioration is evident both in baseline models and in those subject to some form of parameter tuning, reinforcing the case for out-of-sample evaluation as the appropriate standard for model selection in this setting.

The estimators we consider span both linear and regularized paradigms. The expanding window uses a base training window of 40 observations. Standard OLS serves as

a natural benchmark, though it is only recommended when the number of regressors is strictly less than the sample size. We complement it with three penalized linear methods: Lasso, which performs shrinkage and variable selection simultaneously; Ridge, which applies shrinkage without excluding any regressor; and Elastic Net, which nests both as a convex combination.

Relying solely on the naïve R^2 is misleading in this context. Adding regressors mechanically increases in-sample fit and compresses residual variance, which artificially deflates the estimated shock series and may suggest the near-absence of monetary policy shocks altogether. Cross-validation guards against this: reductions in residuals are counted only when they reflect genuine out-of-sample predictive improvement, yielding a more credible decomposition of the policy rate into its systematic and nonsystematic components.

This point has a sharp implication. A model that attributes virtually every change in the policy rate to the systematic component, the part our framework treats as explainable, can always be constructed in-sample, leaving no residual variation to be interpreted as a shock. Out-of-sample discipline is precisely what prevents this degenerate outcome.

Figures 1 and 2 compare the shock series across estimation strategies. The correlation between series is high: despite meaningful differences in out-of-sample performance, the shocks appear visually similar, reflecting the common underlying dataset and methodology shared across specifications. This similarity, however, does not diminish the importance of the cross-validation strategy. Small differences in the estimated shocks can propagate into meaningfully different impulse responses, with direct implications for whether the identified shock is consistent with theoretical predictions.

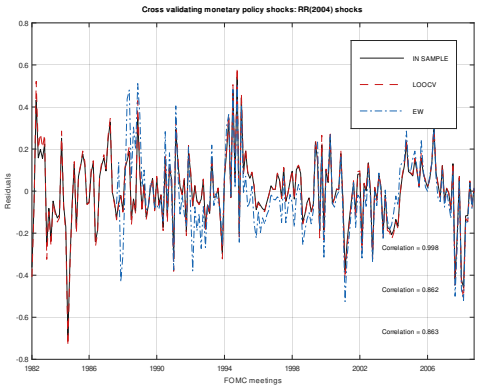


Figure 1: Residual Estimation for Romer and Romer (2004)

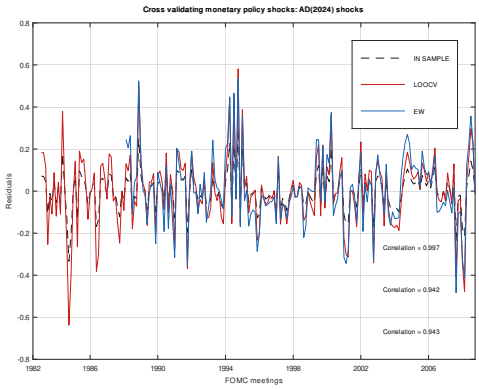


Figure 2: Residual Estimation for Aruoba and Drechsel (2024)

Table 1

	Regressors	R^2	EW R^2	LOOCV R^2
Panel A. Romer and Romer (2004)				
OLS	18	0.499	0.247	0.387
Ridge	18	0.435	0.318	0.368
Elastic Net	18	0.336	0.246	0.309
Lasso	18	0.342	0.251	0.314
Panel B. Expanded set of forecasts				
OLS	133	0.873	-6.081	-0.105
Ridge	133	0.501	0.342	0.382
Elastic Net	133	0.407	0.260	0.324
Lasso	133	0.357	0.249	0.297
Panel C. Forecasts plus sentiments				
OLS	429	1.000	0.216	0.114
Ridge	429	0.675	0.428	0.448
Elastic Net	429	0.490	0.333	0.373
Lasso	429	0.474	0.282	0.359
Panel D. Forecasts plus sentiments plus 4 lags				
OLS	1613	1.000	0.423	0.444
Ridge	1613	0.887	0.416	0.454
Elastic Net	1613	0.561	0.258	0.389
Lasso	1613	0.455	0.288	0.373

Notes: The table reports in-sample R^2, an equally-weighted rolling-window out-of-sample R^2 (EW R^2), and leave-one-out cross-validated (LOOCV) R^2 for the baseline OLS specification and penalized regressions (ridge, elastic net, lasso) across alternative feature sets. Negative out-of-sample R^2 indicates performance below the corresponding mean benchmark. OLS uses the Pseudoinverse (Moore–Penrose inverse) to invert the matrix when the amount of variables exceeds number of observations and provides the Linear Regression which has the minimum L^2 norm among those minimizing the residual sum of squares.

3.5 Effects on the Macroeconomy

In this section, we assess whether the shock series produced by the different cross-validation strategies generates different macroeconomic dynamics. We estimate these dynamic effects using local projections (LP), with our shock series serving as an external instrument for the policy rate to isolate exogenous variation in monetary policy. The choice between VAR and LP involves a bias-variance tradeoff: LP estimates carry lower bias at the cost of higher variance. Crucially, Olea et al. (2025) show that LP is more robust to model misspecification and lag length selection, properties that favor LP when the primary concern is minimizing bias in the estimated impulse responses.

The LP includes seven variables: the one-year Treasury yield (1YTY), industrial production (IP), the unemployment rate (UNEMP), the consumer price index (CPI), a commodity price index (COMMP), the excess bond premium (EBP), and the stock price index (S&P500). The one-year Treasury yield is preferred over the federal funds rate

as the policy indicator because it remained largely unconstrained during the zero lower bound period.

All variables are monthly. The unemployment rate (UNRATE), CPI (CPIAUCSL), and industrial production (INDPRO) are drawn from the FRED database at the Federal Reserve Bank of St. Louis, as is the one-year Treasury yield. The stock price index is gathered from the FRED-MD monthly database also in the Federal Reserve Bank of St. Louis website. The commodity price index corresponds to the monthly CRB CMDT index sourced from Bloomberg. The excess bond premium follows Gilchrist and Zakrajšek (2012) and is obtained from the Federal Reserve Board's website. IP, CPI, and the COMMP enter the LP in logs. This setup closely follows the best practices outlined in Bauer and Swanson (2022). The IRFs are estimated over the period 1984:02–2025:05 with 12 lags.[7]

The instrument is the shock series constructed. It spans February 1982 to October 2008 and is defined at FOMC meeting frequency. To use it as a first-stage instrument for the monthly policy variable, we assign its value to the month of the corresponding FOMC meeting and set it to zero in months without a meeting, a procedure standard in the literature. Instrument strength is assessed via first-stage F-statistics and the instrument-related explained variance,i.e., the reliability measure.

Figure 3 shows that LOOCV yields results qualitatively similar to the in-sample estimates, which is expected given the relatively small number of predictors in this specification. The expanding window results, however, diverge more noticeably, reflecting two compounding forces: the reduced effective sample size in early windows and the lookahead bias that inflates in-sample performance when the full dataset is used for estimation.[8]

In particular, Figure 3 also shows that cross-validation yields different results, as the EW shock is now significant in reducing both industrial production and commodity prices. Moreover, despite greater estimation uncertainty (i.e., wider confidence bands), the effects of this shock on almost all variables are more persistent. Finally, we observe a significant price puzzle in the very short horizon, although it reaches a similar negative peak around 8 months.

[7]Bauer and Swanson (2022) use the two-year Treasury yield instead. All results are robust to using the two-year rate; however, we adopt the one-year rate as our baseline specification due to the stronger first-stage statistics.

[8]We note that the instrument based on the original Romer and Romer (2004) is classified as weak according to the conventional threshold of 10 for the first-stage F-statistic. In addition, we note that both instruments built using cross-validation strategies display even lower first-stage F-statistics and are therefore also weak. Thus, inference based on HAC confidence intervals may be unreliable in the presence of weak instruments.

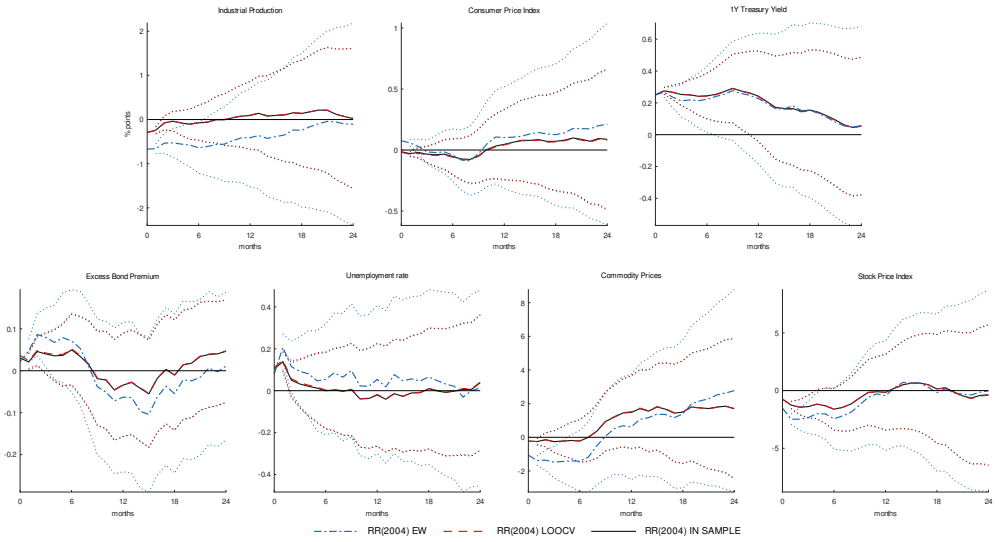


Figure 3: Impulse Response Function using Local Projections for Romer and Romer (2004)

Notes: Dashed lines represent 90% HAC confidence intervals. The identification only uses data that are commonly available for the selected instrument and the endogenous variables. First-stage F-statistics: Romer and Romer (2004) IN SAMPLE: 4.38; Romer and Romer (2004) LOOCV: 4.05; Romer and Romer (2004) EW: 3.27; Reliability: Romer and Romer (2004) IN SAMPLE: 0.09; Romer and Romer (2004) LOOCV: 0.08; Romer and Romer (2004) EW: 0.18;

These findings are corroborated by Figure 4, where analogous discrepancies emerge across cross-validation strategies. Regarding the similarities between the in-sample shock and the LOOCV results, we observe less resemblance than in Figure 3, likely due to the larger set of predictors. Specifically, the EW shock is able to generate theory-consistent impulse responses for CPI and COMMP, although these are only significant at very short horizons. Finally, we also find stronger impact responses for almost all variables.

In summary, the choice of validation procedure is consequential enough to reverse the sign of the point estimates for CPI and commodity prices, underscoring that cross-validation is not merely a diagnostic tool but a substantive identification choice. We advocate that cross-validating is not only crucial to the identification of monetary policy shock but also mandatory in a environment where the number of variables is substantial, which highlights a more robust and precise measurement of the Fed information set.

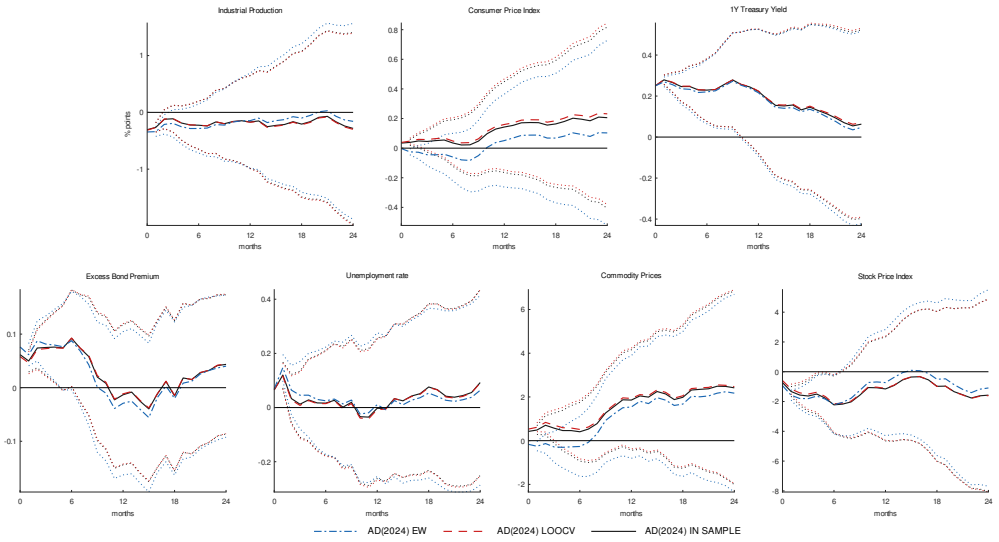


Figure 4: Impulse Response Function using Local Projections for Aruoba and Drechsel (2024)

Notes: Dashed lines represent 90% HAC confidence intervals. The identification only uses data that are commonly available for the selected instrument and the endogenous variables. First-stage F-statistics: Aruoba and Drechsel (2024) IN SAMPLE: 11.61; Aruoba and Drechsel (2024) LOOCV: 12.20; Aruoba and Drechsel (2024) EW: 12.78; Reliability: Aruoba and Drechsel (2024) IN SAMPLE: 0.21; Aruoba and Drechsel (2024) LOOCV: 0.22; Aruoba and Drechsel (2024) EW: 0.30;

4 Large Language Models Shock Estimation

4.1 Large Language Model Embeddings

In the previous section, we examined how the choice of residual estimation method can lead to an overstatement of the Fed information effect. We now turn to the opposite concern: ways in which the literature may have underestimated this effect by relying on a restricted information set. We address this by expanding the information set using Large Language Models (LLMs) and nonlinear estimation methods.

Advances in Natural Language Processing (NLP) and machine learning over the past decade have greatly expanded the tools available for extracting information from text, most notably through LLMs. Relative to classical dictionary-based approaches, LLMs offer three key advantages. First, they are multipurpose: while a dictionary is designed for

a specific task, e.g. the Loughran-McDonald dictionary for financial sentiment analysis, LLMs generate dense vector representations (embeddings) of text that can be used flexibly for downstream classification or regression tasks. Second, LLMs are context-sensitive; the embedding of a word or sentence is shaped by its surrounding text, enabling the model to capture nuanced meaning and generalize to new expressions and vocabulary. Third, LLMs have demonstrated consistently superior performance on both classification and regression tasks across a wide range of domains and applications (Kirtac and Germano (2024), Than et al. (2025), Klähn et al. (2025)).

Despite these advantages, it is not immediately obvious how to apply LLMs to the Greenbook and Beige Book texts. One natural approach is sentiment analysis, either over full sections or at the word and sentence level, to generate scalar classifications. Another possibility, inspired by Vafa et al. (2025), is to use foundational models to produce predictions of target quantities directly relevant to the Fed information effect. Both strategies ultimately require selecting a meaningful dimension of the text, such as sentiment or a specific forecast target, to reduce dimensionality and make linear regression a feasible strategy.

Our approach avoids this compression. We embed the full text using an LLM and incorporate these representations directly into the Fed's information set, without projecting onto an index. This eliminates the information loss inherent in choosing a single summary dimension, at the cost of working with high-dimensional embeddings.

Implementing this approach requires selecting both an embedding model and a machine learning estimator suited to the structure of our data. Several factors govern the performance of LLM embeddings: the domain of the pretraining corpus (Tang and Yang (2025)), the maximum token length the model accepts, and the dimensionality of the resulting embedding space. Treating this as a full combinatorial search, estimating every ML model on every possible embedding, is computationally prohibitive. We therefore proceed in two steps, evaluating embedding models first and estimation methods second.

For the embedding comparison, we consider five models: FinBERT (Araci (2019)), RoBERTa (Liu et al. (2019)), MPNet (Song et al. (2020)), MiniLM, and BGE-Small (Xiao et al. (2023)). These models span a range of pretraining domains and architectures, from finance-specific to general-purpose, and vary in embedding dimension and context length. For each embedding, we estimate Lasso and Ridge regressions of the interest rate change on the embedded text representations, using both LOOCV and an EW to evaluate out-of-sample fit.

Table 2 reports the results of this comparison alongside the embedding dimension of each model. Under Lasso, FinBERT performs best on both the EW and LOOCV criteria; under Ridge, it again yields the best EW result and a near-identical LOOCV result to

the next-best model.

Table 2

	K	R^2	EW R^2	LOOCV R^2
Panel A. Ridge regression				
BGE-Small	518	0.723	0.364	0.462
FinBERT	902	0.520	0.446	0.456
RoBERTa-Base	902	0.621	0.332	0.452
MiniLM	518	0.500	0.300	0.458
MPNet-Base	902	0.611	0.332	0.458
Panel B. Lasso regression				
BGE-Small	518	0.505	0.322	0.416
FinBERT	902	0.465	0.390	0.419
RoBERTa-Base	902	0.445	0.335	0.345
MiniLM	518	0.454	0.273	0.346
MPNet-Base	902	0.481	0.314	0.383

Notes: The table reports in-sample R^2, expanding window (EW) out-of-sample R^2, and leave-one-out cross-validated (LOOCV) R^2 for ridge (Panel A) and lasso (Panel B) regressions using full-text embeddings from five pretrained language models. K denotes the number of features. The optimal penalization parameter α in Panel A is selected via cross-validation.

As discussed above, the pretraining domain is a significant determinant of embedding quality (Tang and Yang, 2025). FinBERT is pretrained on a large corpus of financial news, earnings reports, and analyst commentary (Araci, 2019), making its token representations natively sensitive to the vocabulary and phrasing of central bank communication. General-purpose models such as RoBERTa and MPNet are pretrained on broad web corpora where financial and monetary policy language is a small fraction of the training distribution. As a result, expressions specific to FOMC deliberations are likely to receive more semantically precise representations under FinBERT than under models that have encountered them only incidentally. We therefore adopt the FinBERT embedding for all subsequent analysis.

4.2 What Can Be Learned From LLM Embeddings?

Since textual data structures processed by NLP algorithms are often difficult to interpret, we aim to describe what can be learned and inferred from our FinBERT embeddings. We assess the extent to which our LLM embeddings capture the essence of the communication conveyed in FOMC documents. We also examine how the language inherent in these documents evolves over time and whether it remains stable across periods. In addition, we investigate the relationship between the Fed Chair and the communication strategy adopted during their tenure, as well as potential differences across presidential administrations.

We begin by examining how the linguistic structure of Greenbook communications evolves over time. Figure 5 plots the cosine distance between the embeddings of each meeting and its immediate predecessor, with key economic events annotated. Two features stand out. First, consecutive-meeting distances remain small throughout most of the sample, reflecting the Fed's commitment to policy continuity and gradualism, abrupt linguistic shifts are the exception rather than the rule. Second, the largest spikes coincide precisely with moments of macroeconomic disruption: the 9/11 emergency cut of September 2001, the geopolitical uncertainty surrounding the Iraq War buildup in late 2002, and the Bear Stearns collapse in March 2008. This suggests that embedding volatility carries genuine economic content, spiking when the Fed is forced to reassess its outlook rather than simply updating prior forecasts.

Figure 6 offers a complementary perspective by measuring the cosine distance of each meeting's embedding relative to a fixed reference point: the meeting of August 21, 1990, the first FOMC meeting convened after the onset of the Gulf War and the beginning of the 1990–91 recession. While the high-frequency pattern of consecutive distances remains small, consistent with gradualism, the long-run distance from this reference grows persistently over time. This divergence reflects the cumulative effect of small linguistic adjustments that, accumulated across decades, produce fundamentally different communication styles across monetary regimes. Taken together, the two figures illustrate an important distinction: the Fed communicates consistently in the short run, yet its language undergoes substantial structural drift in the long run, with the embedding space faithfully capturing both dimensions of this evolution.

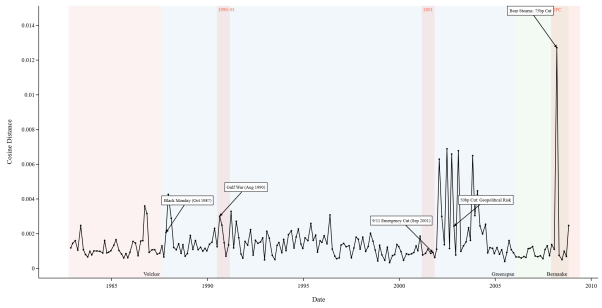


Figure 5: Cosine Distance between Successive FOMC Meetings

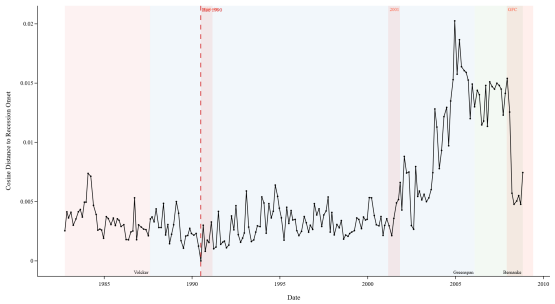


Figure 6: Cosine Distance of each meeting to the closest meeting after the day Iraq invaded Kuwait

To characterize the structure of the embedding space, we employ a set of complementary dimensionality reduction and classification exercises. We begin with t-distributed Stochastic Neighbor Embedding (van der Maaten and Hinton, 2008), a nonlinear projection that maps the high-dimensional embeddings onto two dimensions by minimizing the Kullback-Leibler divergence between the original pairwise similarity distribution and its low-dimensional counterpart, given a perplexity parameter governing the effective

neighborhood size. Because t-SNE preserves local neighborhood structure rather than global distances, we interpret cluster membership and compactness as informative, while treating inter-cluster distances and axis scales as non-structural.

Figures 7–10 present four views of the same t-SNE projection. Figure 7 reveals pronounced temporal dependence in the embedding space: meetings cluster tightly by era, suggesting that FinBERT captures systematic shifts in the vocabulary and tone of Federal Reserve communications over time. Figures 8 and 9 color the same projection by Federal Reserve Chair and sitting President, respectively. Both reveal regime-level clustering, consistent with the hypothesis that communication style varies systematically across chairmanships and, to a lesser extent, across political administrations. Figure 10 augments the chairmanship plot with edges connecting consecutive meetings in chronological order, making the directional drift of Fed language across regimes explicit. We note that t-SNE results are sensitive to the choice of perplexity and initialization; accordingly, we treat these figures as descriptive evidence and complement them with formal statistical tests below.

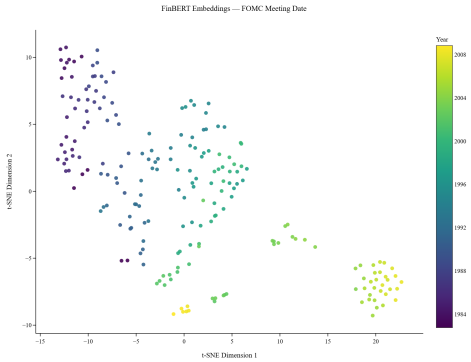


Figure 7: FinBERT Embedding T-SNE by Date

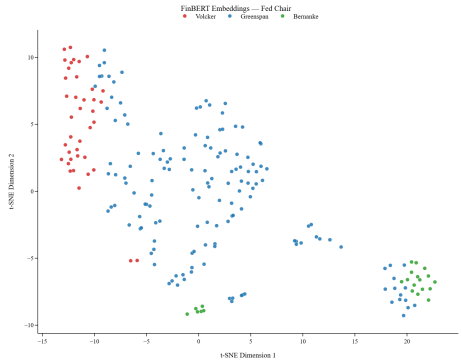


Figure 8: FinBERT Embedding T-SNE by FED Chair

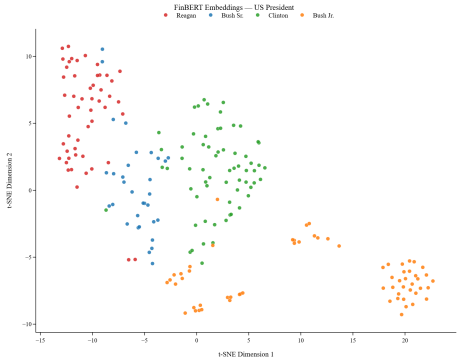


Figure 9: FinBERT Embedding T-SNE by President

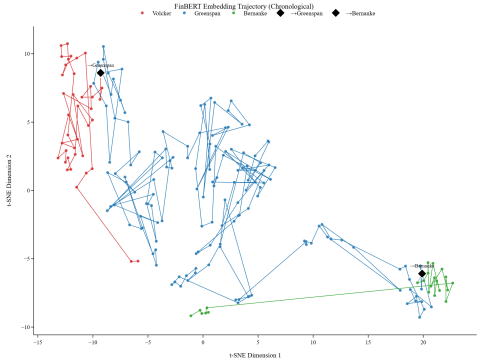


Figure 10: FinBERT Embedding T-SNE by FED Chair with Trajectories

To quantify regime separability, we pursue two complementary approaches. First,

we fit a Support Vector Machine with a radial basis function kernel on the 50 leading principal components of the embedding matrix and evaluate out-of-sample accuracy via five-fold cross-validation, reporting results against the majority-class baseline. Second, we compute the mean pairwise cosine similarity among all meetings within each chairmanship and presidential term as a measure of intra-regime linguistic cohesion, and test for differences across regimes using Mann-Whitney tests.

Figures 11 and 12 report the SVM results, with decision boundaries visualized in the two-dimensional PCA space for illustration; the reported accuracy is estimated in the full 50-dimensional space. Classifying by Federal Reserve Chair yields a cross-validated accuracy of 0.803, against a majority baseline of 0.708, a modest but meaningful improvement that reflects the dominance of the Greenspan era in the sample. The classification by presidential administration is considerably more accurate, achieving 0.789 against a majority baseline of 0.306. We caution, however, that this result should not be interpreted only as direct evidence of political influence on Federal Reserve communication: presidential terms and chairmanships overlap substantially in the sample, so the president classifier also recovers the specific linguistic structure identified in the chairmanship analysis.

Figures 13 and 14 display the distribution of pairwise cosine similarity across all unique meeting pairs within each regime r, Table 3 reports the corresponding Mann-Whitney U tests for distributional differences across regimes. The chairmanship results in Figure 13 are consistent with the patterns observed in the t-SNE analysis. The Greenspan era exhibits the widest dispersion, reflecting both the length of his tenure, which spans multiple distinct macroeconomic episodes, including the dot-com expansion, the September 2001 shock, and the early 2000s recovery, and a communication style that evolved substantially over nearly two decades. By contrast, the Volcker and Bernanke eras display considerably tighter distributions, consistent with more homogeneous communication within each chairmanship. The Volcker era cohesion likely reflects the singular focus on disinflation that characterized the early 1980s, while the Bernanke era cohesion plausibly captures the internally consistent vocabulary of crisis management and forward guidance developed during the Global Financial Crisis. The Mann-Whitney tests confirm that the Greenspan distribution is significantly different from both Volcker and Bernanke. The absence of a statistically significant difference between Volcker and Bernanke is itself informative: despite operating in very different macroeconomic environments, both chairs appear to have adopted comparably focused and internally consistent communication strategies, suggesting that regime clarity produces similarly cohesive embedding structure.

Figure 14 presents the analogous decomposition by presidential administration. Intra-regime similarity is generally higher and distributions are tighter than in the chairmanship analysis, which is consistent with the longer and more stable political cycles relative to

the macroeconomic cycles that tend to define chairmanship regimes. The notable exception is the Bush Jr. administration, which exhibits markedly wider dispersion. This is probably attributable to the diverse macroeconomic conditions that characterized that period, spanning the post-dot-com stabilization, the pre-crisis expansion, and the Global Financial Crisis in 2007-2008, all of which would be expected to produce heterogeneous language in the Greenbooks. The Mann-Whitney tests find statistically significant differences across nearly all presidential pairs, with the single exception of the Reagan-Bush Sr. comparison. This result is consistent with a communication continuity hypothesis: the transition from Reagan to Bush Sr. coincided with the final years of the Volcker chairmanship under both administrations, and the Fed's language during this period was likely driven more by the own monetary policy concerns than by any shift in the political environment. The result therefore reinforces the broader interpretation that era-specific linguistic structure in the Greenbooks is primarily anchored by own monetary policy discussions rather than the political cycle.

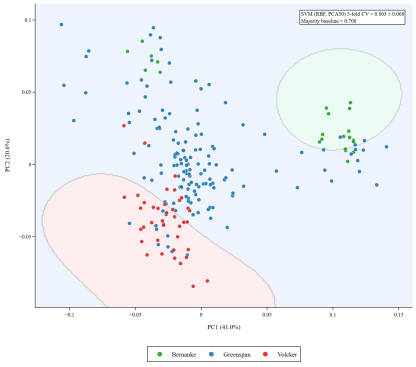


Figure 11: SVM Decision Boundary by FED Chair

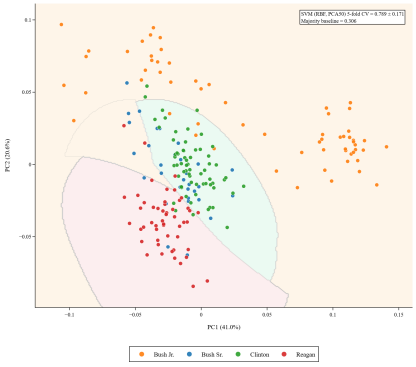


Figure 12: SVM Decision Boundary by President

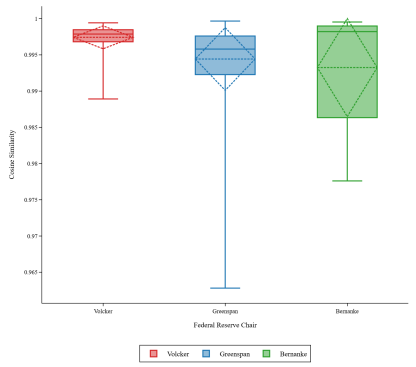


Figure 13: Boxplot for Cosine Similarity Distribution by FED Chairman

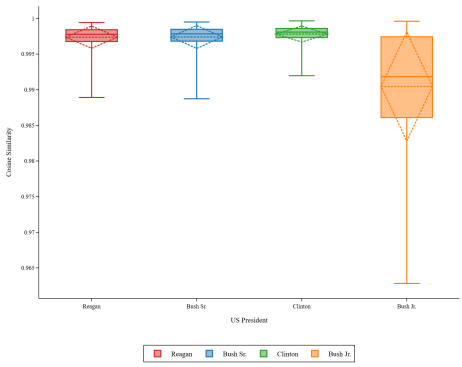


Figure 14: Boxplot for Cosine Similarity Distribution by President

Table 3: Pairwise Mann-Whitney U Tests: Embedding Cohesion

Panel A: By FED Chair				*Panel B: By President*			
Chair A	Chair B	U Stat.	p-val	Pres. A	Pres. B	U Stat.	p-val
Volcker	Greenspan	6,040,220	<0.001	Reagan	Bush Sr.	306,910	0.336
Volcker	Bernanke	88,267	0.472	Reagan	Clinton	1,073,768	<0.001
Greenspan	Bernanke	1,093,205	<0.001	Reagan	Bush Jr.	1,937,100	<0.001
				Bush Sr.	Clinton	434,808	<0.001
				Bush Sr.	Bush Jr.	757,530	<0.001
				Clinton	Bush Jr.	3,179,799	<0.001

Notes: Mann-Whitney U test. H_0: equal within-group cosine similarity distributions. A p-value <0.05 rejects equal cohesion distributions.

4.3 Machine Learning Residual Estimation

Having selected the embedding model, we turn to the choice of estimator. We use the extended set of forecasts described at Section 3.5 and consider four model classes: linear regularized models (Ridge and Elastic Net), Random Forests (RF), Support Vector Machines (SVMs), and XGBoost. Though deliberately limited in scope, these classes span a wide range of complexity and inductive biases, ensuring that no single modeling paradigm dominates the comparison by design.

Given the small sample size, linear regularized methods, Ridge and Elastic Net, provide a natural benchmark. Their bias-variance tradeoff is well understood, and their performance in low-observation, high-dimensional settings is well established in the literature. We complement these with two nonlinear approaches. Random Forests and XGBoost are tree-based ensemble methods well suited to capturing complex interactions among predictors. SVMs offer a distinct alternative: by mapping inputs into a high-dimensional feature space via a kernel function, they can recover nonlinear decision boundaries while remaining robust to overfitting through margin maximization.

Finally, we implement two-stage procedures using Lasso. In the first stage, the full model is estimated and predictors with zero coefficients are discarded. In the second stage, the model is fitted on the selected subset. This sequential approach sharpens feature selection and can improve out-of-sample performance, particularly in high-dimensional settings where many predictors carry little signal.

Table 4 reports the best out-of-sample estimates for each ML method. Among the remaining specifications, results are broadly similar and robust across models. XGBoost on the full feature set achieves the best LOOCV performance, while Random Forest yields the best expanding window result by a negligible margin. We therefore adopt the XGBoost-generated series as our shock measure for the local projection analyses in the following section.

Table 4

	N	R^2	EW R^2	LOOCV R^2
Panel A. Stage 1 — All features				
Ridge	210	0.477	0.437	0.434
Elastic Net	210	0.330	0.342	0.386
Random Forest	210	0.760	0.450	0.462
SVM	210	0.572	0.449	0.434
XGBoost	210	0.956	0.449	0.501

Notes: Each model is fitted directly on the full embedding feature set ($K = 902$) and tuned via cross-validation. $N = 210$ observations in all cases. EW R^2 and LOOCV R^2 are out-of-sample metrics.

The two-stage specifications perform considerably worse than their single-stage counterparts (as can be seen in Table 5), a result driven by the aggressive variable selection of the first-stage Lasso, which retains only five embedding dimensions and real GDP growth. Although at the cost of predictive performance, the variable selection form Lasso is in itself very informative of the relative importance of the embedding variables when compared to the Greenbook forecasts.

Table 5

	N	R^2	EW R^2	LOOCV R^2
Panel B. Stage 2 — Lasso pre-selection				
Ridge	210	0.540	0.406	0.467
Elastic Net	210	0.537	0.401	0.466
Random Forest	210	0.635	0.423	0.466
SVM	210	0.516	0.409	0.467
XGBoost	210	0.865	0.373	0.460

Notes: Two-stage procedure in which Lasso first selects 7 features (`embed_177`, `embed_316`, `embed_415`, `embed_587`, `embed_720`, `rgdp0`, `rgdp0d`), and each model is then re-estimated on the selected subset. All models are tuned via cross-validation. $N = 210$ in all cases. EW R^2 and LOOCV R^2 are out-of-sample metrics.

To assess the marginal informational content of the text embeddings relative to the Greenbook forecasts, we estimate a Random Forest model under three feature configurations: macroeconomic variables alone, text embeddings alone, and the two combined. Variable importance is evaluated via two complementary measures. The Mean Decrease in Impurity (MDI) quantifies the share of total variance attributed to each predictor across all splits in the ensemble. The Mean Decrease in Accuracy (MDA) measures the reduction in out-of-bag predictive accuracy induced by randomly permuting a single feature's values, holding all others fixed (Breiman, 2001; Louppe et al., 2013).

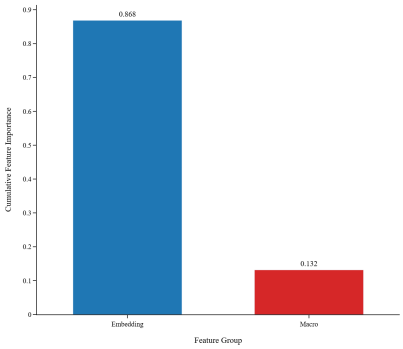


Figure 15: Variable Importance by MDI

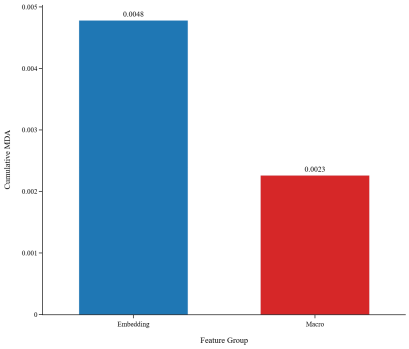


Figure 16: Variable Importance by MDA

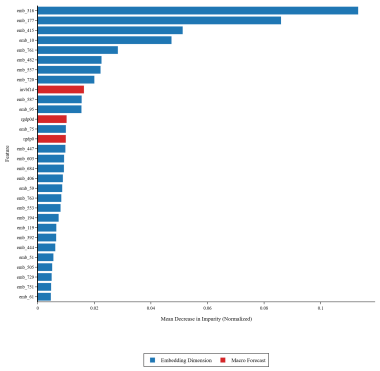


Figure 17: 30 Most Important Variables by MDI

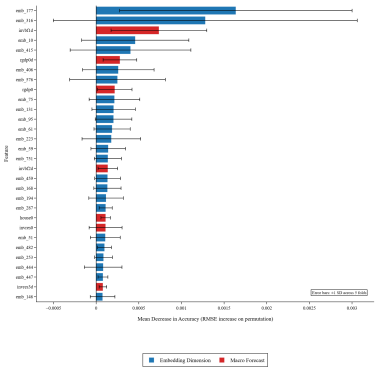


Figure 18: 30 Most Important Variables by MDA

Table 6 shows that the embedding feature set yields lower cross-validated RMSE and higher R^2 than the macroeconomic variables alone, and that combining both sources produces the strongest performance across all metrics. Figures 15 and 16 confirm that the aggregate importance of the embedding group, measured by both MDI and MDA, is at least twice that of the macro group. Figures 17 and 18 further reveal that individual embedding features account for a disproportionate share of the top-ranked predictors.

Table 6

	CV RMSE	RMSE Std	CV R^2	R^2 Std
Macro	0.1826	0.0130	0.3433	0.1972
Embedding	0.1784	0.0093	0.3899	0.1199
Macro + Embedding	0.1704	0.0090	0.4413	0.1223

Notes: CV RMSE and CV R^2 are cross-validated metrics. Std denotes the standard deviation across folds.

4.4 Effects on the Macroeconomy: LLM Based Shocks

Given our two LLM-based shocks, we aim to understand what information they convey. We begin with a visual inspection and identify the five meetings showing the largest shock magnitudes. Figure 19 shows the behavior of both LLM shock series. The correlation between them is approximately 0.90. Although the shocks appear visually similar, there are some notable exceptions: (i) years in which the shocks move in opposite directions—1988, 1993–1994, and 2001; and (ii) years in which the shocks differ in magnitude, particularly toward the end of the sample, during periods encompassing the dot-com bubble and the Global Financial Crisis.

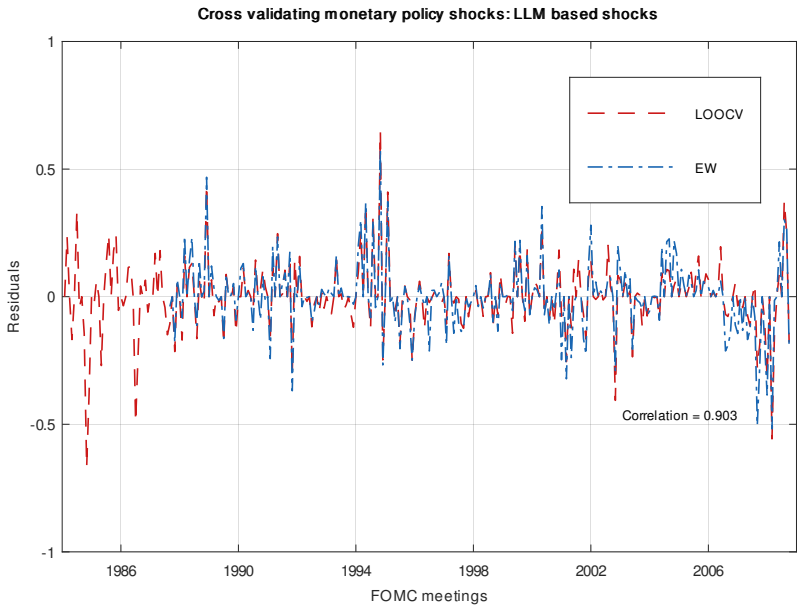


Figure 19: Cross validating monetary policy shocks: LLM based shocks

Table 7 presents the largest values of our shock series, considering both positive and negative values. It is important to note that positive (negative) values correspond to contractionary (expansionary) surprises. According to our procedure, four of the five meetings with the highest positive shocks are common across both cross-validation strategies. The contractionary extreme meetings, December 1988, May and November 1994, and February 1995, correspond to well-documented tightening cycles. The 1994–1995 episode in particular is one of the most studied in the monetary policy literature, as the Fed raised rates aggressively in a short window, catching markets off guard.

On the expansionary side, the asymmetry between the two series is more pronounced. The LOOCV series flags episodes from the mid-1980s (November and December 1984, July 1986) and the post-dot-com easing (November 2002), while the EW series concentrates its extreme negative values in the 2001–2008 window. This divergence likely reflects how each strategy weights earlier observations. The EW approach, which begins

only after September 1987, naturally omits the mid-1980s easing cycle entirely. March 2008 stands as the only common extreme negative shock, capturing the Fed's aggressive emergency cut following the Bear Stearns collapse, a landmark expansionary surprise that both series robustly identify.

Table 7

Date	Shock	Date	Shock
Panel A. LOOCV LLM Shock			
12/1988	0.489	11/1984	-0.670
05/1994	0.358	12/1984	-0.379
11/1994	0.650	07/1986	-0.493
02/1995	0.410	11/2002	-0.406
08/2008	0.380	03/2008	-0.557
Panel B. EW LLM Shock			
12/1988	0.468	11/1991	-0.369
05/1994	0.367	02/2001	-0.322
11/1994	0.574	09/2007	-0.504
02/1995	0.372	01/2008	-0.385
05/2000	0.353	03/2008	-0.518

We now assess whether the shock series constructed under our approach generates macroeconomic dynamics consistent with the theoretical predictions for monetary policy shocks. The LP specification, data sources, and instrumentation procedure are as described in Section 3. The key difference here is the instrument: we use the series selected on the basis of its superior out-of-sample accuracy under both the LOOCV and expanding window criteria. Theory predicts that a contractionary shock should compress output, the price level, and commodity prices; raise the unemployment rate, the policy indicator, and the excess bond premium.

Figure 20 presents the IRFs for a monetary policy shock of 25 basis points (bp). For reference, we include the shocks identified by Bauer and Swanson (2023), Miranda-Agrippino and Ricco (2021), and Aruoba and Drechsel (2024). The two LLM-based series are shown as the LOOCV shock and the EW shock, with dotted lines of the same colors denoting the corresponding 90% confidence intervals. [9]

By construction, both LLM-based shocks produce an increase in the one-year Treasury yield, with a magnitude normalized to 25bp on impact. These responses are very persistence lasting more than 10 months no matter the instrument considered. A contractionary shock generates an approximately 3 bp decline in industrial production on impact, implying an economic downturn that lasts roughly one year; however, the effects are statistically significant only at short horizons, with the exception of Bauer and Swan-

[9] All instruments pass the conventional first-stage F-statistic threshold of 10 (except for the LLM-LOOCV) and exhibit reliability measures comparable to those of high-frequency surprises.

son (2023). The unemployment rate response is uniform across instruments: an initial rise that decays back toward steady state after just a few months.

The most notable differences emerge for commodity prices, the consumer price index (CPI), and the excess bond premium (EBP). Only the EW shock generates a decline in commodity prices. For CPI, the LOOCV shock fails to resolve the price puzzle, while the EW shock produces a delayed negative response emerging after six months.[10] Similarly, only the EW shock produces the expected tightening in financial conditions, as measured by a rise in the EBP. Although the EW shock delivers superior point estimates, its higher variance renders most results statistically insignificant, highlighting potential gains from dataset expansion. The high-frequency shocks of Miranda-Agrippino and Ricco (2021) and Bauer and Swanson (2023) align more closely with theoretical predictions across these margins.

In terms of magnitude, our results are broadly consistent with the literature. The impulse responses display qualitative similarities with Jarociński and Karadi (2020), Miranda-Agrippino and Ricco (2021), Bauer and Swanson (2022), and Aruoba and Drechsel (2024), and these patterns remain robust to alternative LP specifications, sample choices and instruments used.[11]

[10]The benchmark narrative approach exhibits even greater delay, with price responses turning negative only after 18 months in some cases (Aruoba and Drechsel, 2024).

[11]Again, all results are robust to using the two-year Treasury yield; however, we adopt the one-year rate as our baseline specification due to the stronger first-stage statistics of our proposed instrument.

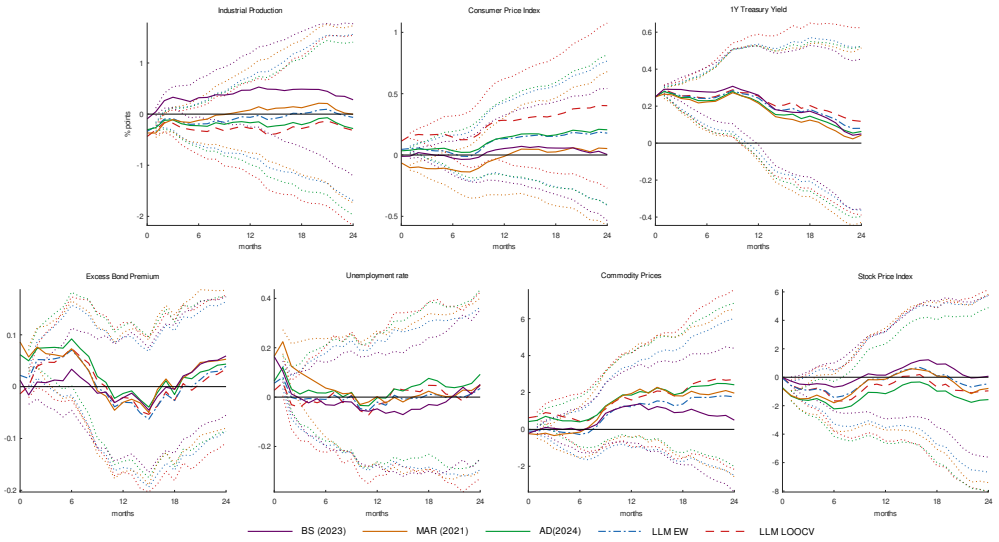


Figure 20: Impulse Response Function using Local Projections for the LLM based shocks

Notes: Dashed lines represent 90% HAC confidence intervals. The identification only uses data that are commonly available for the selected instrument and the endogenous variables. First-stage F-statistics: LOOCV LLM-based: 8.85; EW LLM-based: 10.66; Aruoba and Drechsel (2024): 11.61; Miranda-Agrippino and Ricco (2021):13.98; Bauer and Swanson (2023): 28.28. Reliability: LOOCV LLM-based: 0.20; EW LLM-based: 0.18; Aruoba and Drechsel (2024): 0.21; Miranda-Agrippino and Ricco (2021):0.28 ; Bauer and Swanson (2023): 0.21.

5 Conclusion

This paper seeks to identify monetary policy shocks using textual information from central bank documents related to monetary policy decisions. This task is challenging because it requires overcoming biased estimation and overfitting, problems which this paper addresses by introducing cross-validation methods for residual estimation such as the Leave-one-out croos-validation (LOOCV) and the expanding window (EW) to this literature. Moreover, we provide evidence that cross-validation is not merely a diagnostic tool but a substantive identification choice.

Aiming to provide a broader measure of the Fed's information set, we employ FinBERT embeddings derived from FOMC documents. First, our LLM embeddings expand the notion of this information set, with results that remain robust even when evaluated using out-of-sample metrics. This measure is not a mere coincidence, as we provide evidence that the embeddings capture macroeconomic information and have predictive

power for the interest rate target.

Regarding the effects of our constructed shock measures on the macroeconomy, under the narrative approach, we provide evidence that they are consistent with theoretical predictions. Moreover, our results help narrow the gap between the narrative and high-frequency approaches. It is important to note that these two identification strategies rely on fundamentally different foundations, yet share the same objective of isolating exogenous variations in monetary policy.

A separate and recently emphasized concern is that applications using large language models (LLMs) may be exposed to an additional form of look-ahead bias. Levy (2024) and Lopez-Lira et al. (2025) note that the pretraining corpora of widely used LLMs can contain information published after the period under study. As a result, even when the econometric design enforces strict out-of-sample evaluation, the text-based features produced by an off-the-shelf LLM could, in principle, reflect information that was not available at the time.

Recent work provides a potential path forward. He et al. (2025) propose training and deploying LLMs under chronological constraints, restricting both pretraining and fine-tuning data to information available up to a given cutoff date. Implementing chronologically consistent LLM pipelines for central bank communication is beyond the scope of this paper, but it represents a promising direction for future research. Moreover, reconciling our broader measure of the Fed's information set, via high-dimensional LLM embeddings, with high-frequency identification strategies offers a particularly fruitful avenue for subsequent work.

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